

Computer Interfacing

Section-A

Compulsory → slide 1 + 2

1st slide

- i) why computer interfacing needed
- ii) Basic i/o diagram
- iii) Data Transfer Method
- iv) Printer pin diagram (single Handshaking)
↳ waveform with explanation

2nd slide

- 82C55 ***
- ↳ Interfacing with μ -processor with decoder
 - ↳ Command Register
- Mode-0
- Keyboard with Decoder Mechanism
- ↳ Flowchart

3rd slide

- Programmed I/O
- Interrupt ***
- Priority Interrupt, Daisy chain & Nested Int. ***
- ↳ Interrupt handling
 - ↳ Vector Table
 - ↳ Type, Flag

Serial Communication

↳ Math solve with type variation

Asynchronous & Synchronous Transfer

Bit stuffing

RS232 → UART 16550

Register

Code ***

→ Initialization

→ Sending data

→ Receiving data

Error Type

Interfacing with High Power Devices

Stepper Motor (full step)

Inductive Kick

Darlington Pair

Code

Magnetic Drum Read/Write

Magnetic Disc

Optical Storage

Read/Write Mechanism

Focus Control

Track "

Section-B

Compulsory → DMA

USB slide

↳ Packet type

Topology

Packet type

u u explanation

Ack & Data Packet Loss

Barcode

↳ Explanation

UPC & EAN

Left Hand A, Left Hand B & Right Hand

Barcode conversion

Checksum

DMA

Block Diagram with pin (known pins only)

DMA Registers

Step of programming

Code

Sensor slide

↳ short Note

Embedded System → Block Diagram

D/A Conversion

↳ u-computer interfacing

Parallel A/D

SAR A/D → step by step

SAR Block ^{Diagram} & Explanation

DC Motor Speed Control Error & Solution

(Without Figure)

Printer

↳ Inkjet
↳ Laser

Set Pattern for Section B

DMA → 1 set

DC Motor, Sensor, DAC, ADC - 2 set
USB + Barcode - 1

DMA

DMA Register

step of programming

code

source code

in that code

in that code